Electronics Technical Interview Reviewer

Analog Devices Cavite student intern preparation
Interview October 20 2026 at 8 AM Philippine time

Start with circuit analysis. Your goal is to look at a schematic, explain where current can flow, calculate the important voltages and currents, and choose the correct measurement. A correct explanation matters as much as the final number.

How to use this reviewer

1. Read pages 2 to 9 in order. Redraw every circuit and solve each worked example before looking at the next step.

2. Study components and meters on pages 10 to 17. Say the measurement method and expected reading aloud.

3. Answer the technical questions on pages 18 and 19. Take the practice test on pages 20 and 21, then use the solutions on pages 22 and 23.

4. Use page 24 for final recall. When you miss a question, identify whether the problem was circuit connections, units, signs, a component model, or arithmetic.

The YouTube guide on page 25 follows the same learning order. Watch a topic, pause to solve an example yourself, and then return to the matching practice problem.

For unfamiliar words, use the DC and AC explanation on page 26 and the Definition of Terms section on pages 27 to 30.

Your six day study plan

Date	Focus	Proof that you learned it
Oct 14	Pages 2 to 5	Explain voltage and current; solve series and parallel circuits.
Oct 15	Pages 6 to 9	Solve a mixed circuit and one node equation without notes.
Oct 16	Pages 10 to 12	Choose the meter mode, jack, connection, and explain loading.
Oct 17	Pages 13 to 16	Explain capacitor timing and solve a diode or LED circuit.
Oct 18	Pages 17 to 21	Answer aloud and finish the practice problems.
Oct 19	Pages 22 to 24	Redo missed problems and rehearse a technical interview.

October 20 morning: Review the formulas and meter connections briefly. Use the same solving routine you practiced. The questions here are practice questions for these topics, not a prediction of ADI’s exact interview.

Reading a Circuit

Why A is 5 V and B is 0 V

Voltage always compares two points. A source marked 5 V tells us that its + terminal is 5 V higher than its − terminal. First, we choose the − terminal as our starting point and call it **0 V**. This chosen starting point is the reference or ground.

Ideal means a perfect version used for a simple calculation. Here, an ideal wire has 0 Ω resistance. Its voltage drop is V = I × 0 = 0 V, so points joined directly by that wire have the same voltage. Real wires have a small resistance that we ignore in these beginner problems. An ideal resistor still has its stated resistance; “ideal” does not mean every component has zero resistance.

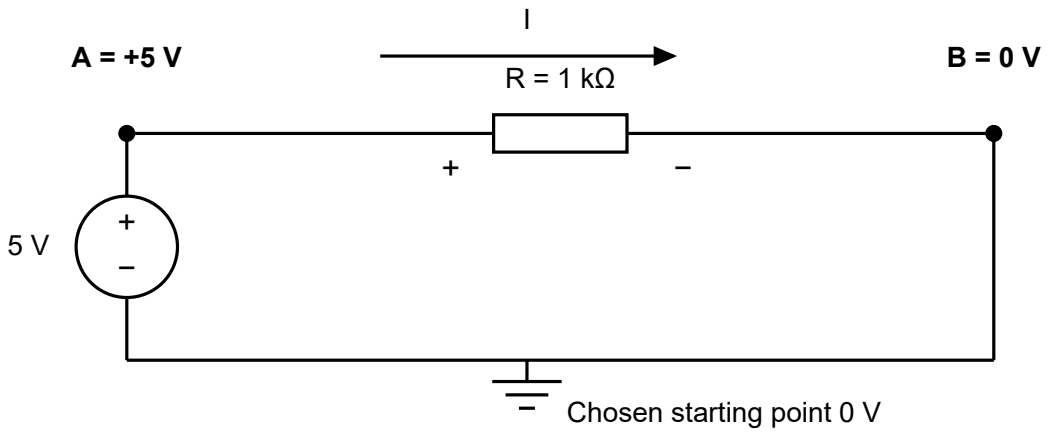


Figure 1. The circle is the 5 V source. The rectangle is resistor R. Follow the wires from A and B back to the source terminals.

Find the point voltages step by step

1. Start at the source − terminal: We chose this point as 0 V. The bottom wire is directly connected to it, so the bottom wire is also 0 V.

2. Follow B: B connects down the right wire and along the bottom wire to the − terminal. There is no resistor along that path. Therefore **B = 0 V**.

3. Follow A: A connects directly to the + terminal. The source raises the voltage by 5 V from − to +, so **A = 0 + 5 = +5 V**.

4. Look across R: Its left side is at 5 V and its right side at 0 V. The difference across the resistor is **5 − 0 = 5 V**. This is the voltage to use in I = V/R.

Voltmeter connection	**Expected reading**
Black probe at source −; red probe at A	+5 V
Black probe at source −; red probe at B	0 V
Black probe at B; red probe at A	+5 V across R

Remember: 0 V does not mean no current. It means the same voltage as the chosen starting point. Also, a source’s − terminal is not automatically 0 V in every circuit; we chose it as 0 V here.

Ohms Law and a Solving Routine

V = IR I = V/R R = V/I

V is the voltage difference between the resistor's two ends. **I** is the current flowing through it. **R** is its resistance. If you know any two, the formulas above give the third. An ordinary fixed resistor follows this rule closely; a diode needs a different rule.

Across versus through: "Across" compares the two ends; "through" follows the path. In Figure 1, 5 V is across R and current flows through R. If there are several resistors, work out the voltage across the particular resistor first.

Units you must get right

Quantity	Base unit	Useful conversion
Voltage V	volt V	1 mV = 0.001 V
Current I	ampere A	1 mA = 0.001 A; 1 μA = 0.000001 A
Resistance R	ohm Ω	1 kΩ = 1000 Ω; 1 MΩ = 1000000 Ω
Capacitance C	farad F	1 μF = 1000 nF; 1 nF = 1000 pF

Fast unit shortcut: V ÷ kΩ gives mA, and mA × kΩ gives V. Example: 5 V ÷ 1 kΩ = 5 mA.

Use this routine on every circuit

1. State what you need to find and the assumptions: DC or changing signal; ideal or real components.

2. Choose the 0 V starting point and label points like A and B. Points joined directly by wire form one node. Look for a single path or separate parallel paths.

3. Choose current arrows and voltage polarities. A negative current means the real direction is opposite your arrow.

4. Simplify valid resistor groups, or write KCL at a node and KVL around a loop.

5. Solve with consistent units. Then check current sums, voltage sums, and whether the answer is physically reasonable.

Worked example 1

1. Figure 1 has 5 V across R. Convert R: **1 kΩ = 1000 Ω**.

2. Use I = V/R: I = 5/1000 = **0.005 A**. Multiply by 1000 to convert A to mA: **5 mA**.

3. Power means how fast the resistor turns electrical energy into heat. P = VI = 5 × 0.005 = **0.025 W = 25 mW**. This is below a 0.25 W rating, if the part's temperature limits allow it.

$P = VI = I^2R = V^2/R$ for a resistor

Predict first: If the voltage doubles and resistance stays fixed, current doubles and resistor power becomes four times larger. If R doubles at the same voltage, current halves.

Resistors in Series

Series means one path. Follow the wire from the source + terminal: current passes through R1, then R2, then returns to the − terminal. There is no place for it to split between R1 and R2, so both carry the same current.

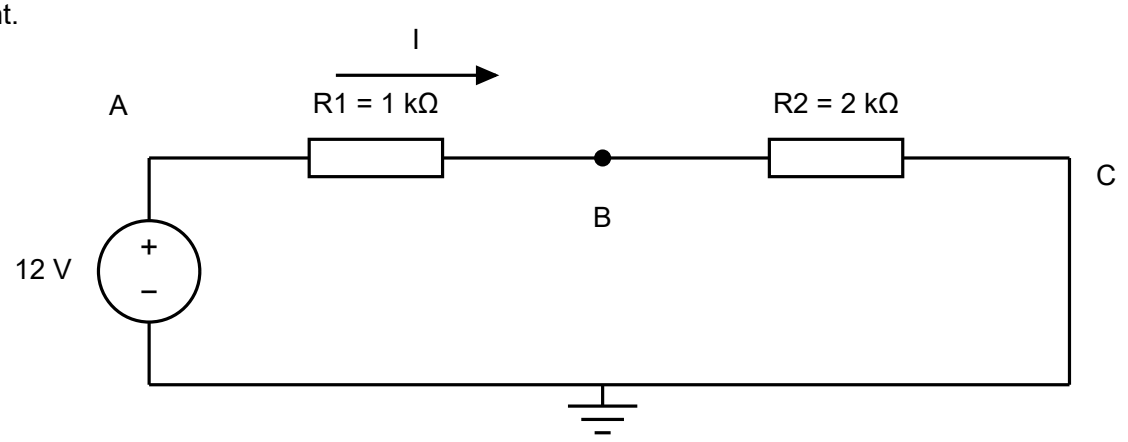


Figure 2. R1 and R2 share node B, and no other branch connects to B.

Rtotal = R1 + R2 + … Same current Voltage drops add

Worked example 2

1. Rtotal = 1 kΩ + 2 kΩ = **3 kΩ**.

2. I = 12 V / 3 kΩ = **4 mA**. This flows through both resistors.

3. VR1 = 4 mA × 1 kΩ = **4 V**. VR2 = 4 mA × 2 kΩ = **8 V**.

4. Choose the source − terminal as 0 V. C connects directly to it, so C = 0 V. A connects to source +, so A = 12 V. Moving across R1 reduces the voltage by 4 V: **B = 12 − 4 = 8 V**. Across R2 it falls another 8 V: 8 − 8 = 0 V at C.

5. P1 = (0.004)² × 1000 = **16 mW**; P2 = **32 mW**. Source power = 12 × 0.004 = 48 mW, equal to total resistor dissipation.

Explain it in an interview

“There is only one current path, so both resistors carry the same current. I add their resistances to get 3 kΩ, find the 4 mA current, and use IR to get the individual drops. The larger resistor has the larger drop because the current is the same.”

Common traps

• Current is not used up after the first resistor. Energy is transferred; steady branch current remains the same.

• You cannot apply 12 V to both resistors separately in this circuit.

• If either resistor opens, the branch current becomes zero. For a single open fault, the open component can have the full source voltage across it.

Try it: A 9 V source supplies 1 kΩ and 2 kΩ in series. Find I and both drops. **Answer:** 3 mA, 3 V, and 6 V.

Resistors in Parallel

Parallel means separate paths between the same two points. Here, the top of each resistor connects to source +. The bottom of each connects to source −. Choose the bottom as 0 V; the top is then 12 V. Each resistor has the same **12 − 0 = 12 V** across it.

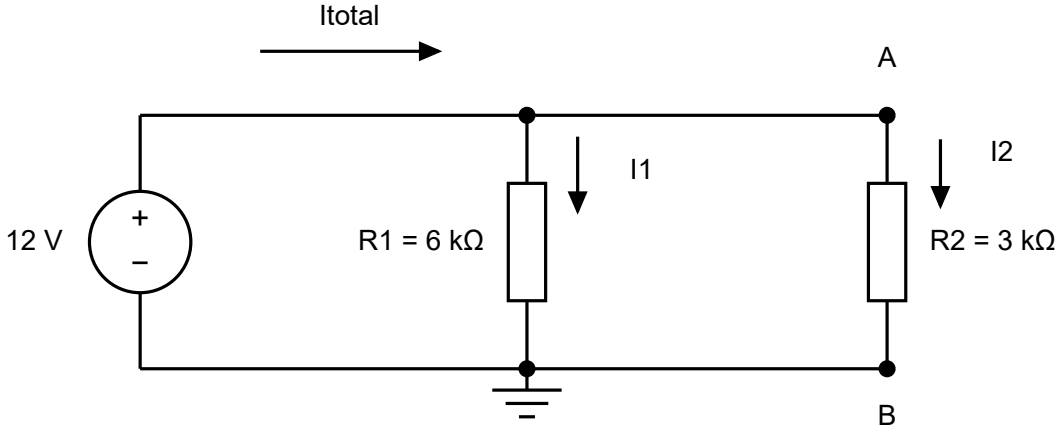


Figure 3. Both resistors connect from A to B. They are parallel even if the drawing is rearranged.

1/Rtotal = 1/R1 + 1/R2 + …

For two resistors Rtotal = R1R2 / (R1 + R2)

Worked example 3

1. Rtotal = (6 × 3)/(6 + 3) kΩ = **2 kΩ**. It is smaller than the smallest branch resistance.

2. Each resistor has **12 V** across it. I1 = 12/6 = **2 mA**; I2 = 12/3 = **4 mA**.

3. Itotal = I1 + I2 = **6 mA**. Check using the equivalent: 12 V / 2 kΩ = 6 mA.

4. P1 = 12 × 0.002 = **24 mW**; P2 = **48 mW**. Source power = 72 mW.

Another way to calculate branch current

For two branches I1 = Itotal × R2/(R1 + R2)

The other branch resistance appears in the numerator. Lower resistance takes more current because both branches see the same voltage. In this example, 6 mA × 3/9 = 2 mA through the 6 kΩ branch.

Equal branches: N equal resistors R in parallel give Rtotal = R/N and equal branch currents. Adding a positive resistance branch lowers Rtotal. At a fixed ideal supply voltage, source current increases.

Try it: 4 kΩ and 4 kΩ are in parallel across 8 V. **Answer:** Rtotal = 2 kΩ, each branch = 2 mA, source current = 4 mA.

Solving a Mixed Circuit

A key skill to practice until the connections feel clear

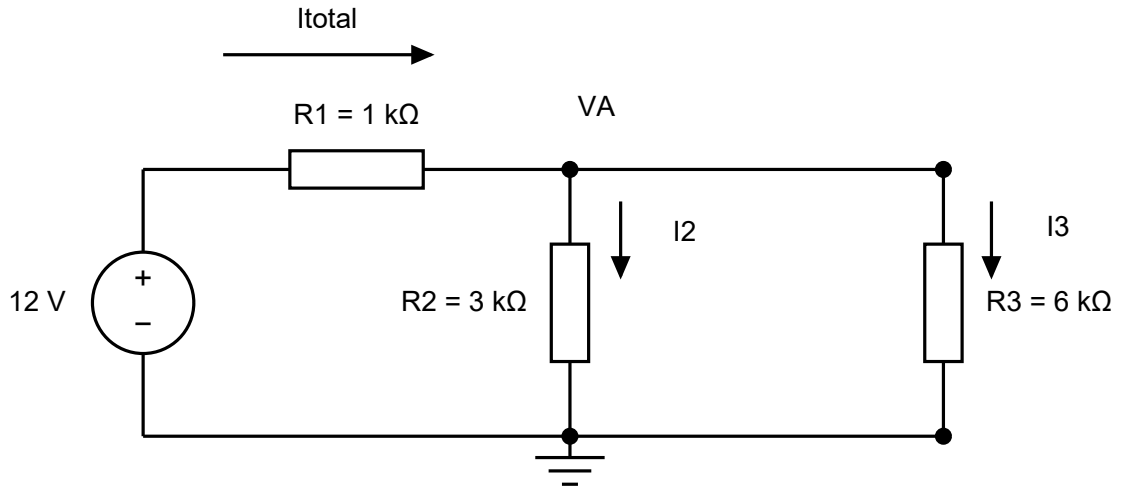


Figure 4. R2 and R3 are parallel. R1 is in series with their equivalent resistance.

Follow the path: current first passes through R1. At the point marked VA, it splits between R2 and R3. Those two paths join again at the bottom. This is why R1 carries the total current while R2 and R3 carry parts of it.

Worked example 4

1. Start at the parallel group: Rp = (3 × 6)/(3 + 6) = **2 kΩ**.

2. Replace that group with one 2 kΩ resistor: Rtotal = 1 + 2 = **3 kΩ**.

3. Source current and R1 current: Itotal = 12/3 = **4 mA**.

4. Voltage across R1: VR1 = 4 mA × 1 kΩ = **4 V**.

5. Voltage left for the parallel group: VA = 12 − 4 = **8 V**. R2 and R3 each have 8 V, not 12 V.

6. I2 = 8/3 = **2.667 mA**; I3 = 8/6 = **1.333 mA**.

7. Check KCL: 2.667 + 1.333 = 4 mA. Check KVL: 12 − 4 − 8 = 0 V.

Why the wrong shortcut fails

R1 and R2 are not in series: current can split into R3 at their shared node. R1 and R3 are not parallel: they do not share both end nodes. First trace connections; only then choose a formula.

Explain your approach

“I will reduce the parallel pair first, calculate the supply current, then work backward to recover the branch voltage and currents. I will verify the branch currents add to the supply current.”

Try it: Keep the 12 V source and 1 kΩ R1, but make R2 = R3 = 4 kΩ. **Answer:** Rp = 2 kΩ, Itotal = 4 mA, VA = 8 V, each branch = 2 mA.

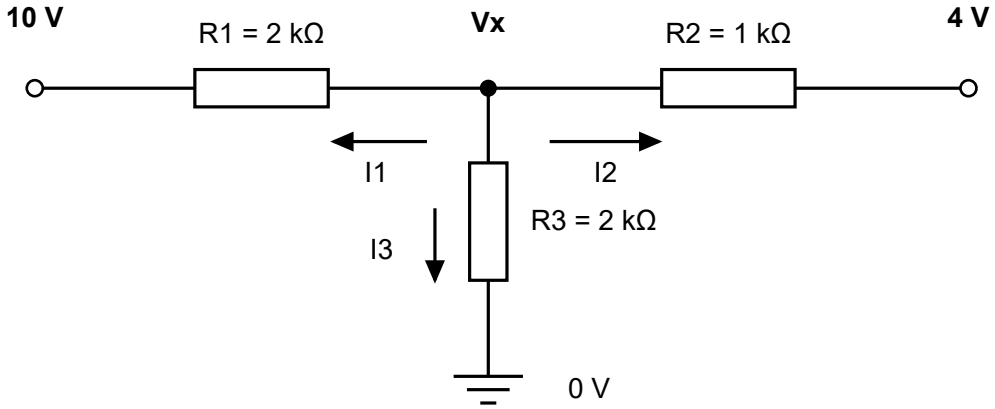
Kirchhoffs Current Law

KCL and one node analysis

KCL is a current balance at a meeting point. A node is a point or connected wire where circuit paths meet. In these circuits, current does not pile up there: total current arriving must equal total current leaving. [2]

ΣI = 0 or total entering = total leaving

A simple junction has 7 mA entering and two currents leaving: 2 mA and Ix. Therefore 7 = 2 + Ix, so **Ix = 5 mA**. If your result is negative, reverse the assumed direction.



Both known voltage sources use this 0 V reference

Figure 5. Vx is connected through resistors to known 10 V, 4 V, and 0 V nodes. Both sources use the same reference.

Worked example 5

Vx means the voltage we do not know yet. The wires on the left, right, and bottom have known voltages: 10 V, 4 V, and 0 V. Resistors lie between them and Vx, so Vx is not automatically any of those values.

For each path, calculate current as **(voltage at start − voltage at end) ÷ resistance**. Draw all three arrows away from Vx and count those currents as positive. The left path gives (Vx − 10)/2; the right gives (Vx − 4)/1; the bottom gives (Vx − 0)/2. Here V divided by kΩ gives mA.

(Vx − 10)/2 + (Vx − 4)/1 + Vx/2 = 0

The three signed currents must add to zero. To remove the fractions, multiply **every term** by 2: Vx − 10 + 2Vx − 8 + Vx = 0. Collect the Vx terms: 4Vx − 18 = 0. Thus 4Vx = 18 and **Vx = 18/4 = 4.5 V**.

Branch	Current with chosen direction	Meaning
Vx to 10 V	(4.5 − 10)/2 = −2.75 mA	2.75 mA enters Vx.
Vx to 4 V	(4.5 − 4)/1 = 0.50 mA	0.50 mA leaves Vx.
Vx to 0 V	4.5/2 = 2.25 mA	2.25 mA leaves Vx.

Check: 2.75 mA entering = 0.50 + 2.25 mA leaving. The negative first current is physically reasonable: current flows from the higher 10 V node toward the 4.5 V node.

Use nodal analysis when: A resistor network does not simplify neatly, or the main unknown is a node voltage. Start with one unknown node before attempting several.

Kirchhoffs Voltage Law

KVL and consistent signs

KVL is a voltage balance around a complete path. Start at the source − terminal, go around the circuit, and return to where you started. The source raises voltage; the resistors reduce it along the current direction. Total rises must equal total drops. [2]

ΣV = 0 or total rises = total drops

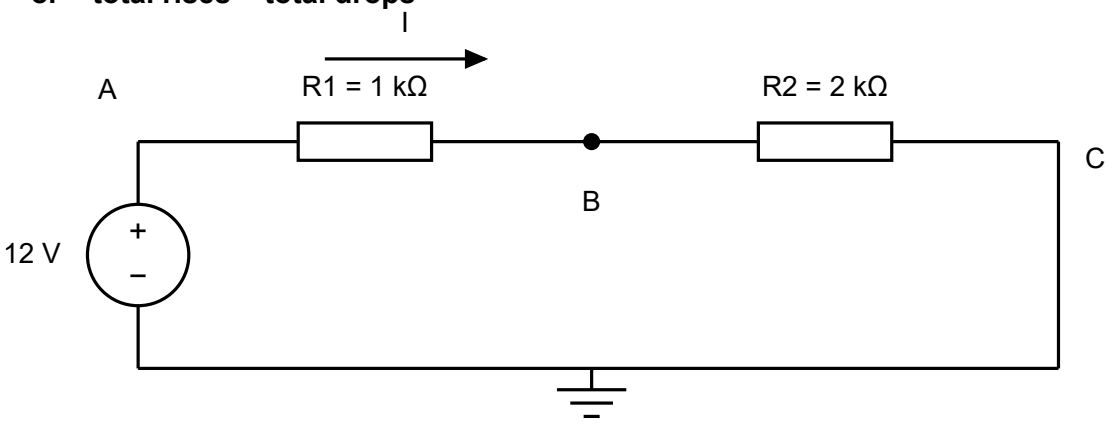


Figure 6. Traverse clockwise from source − through source +, R1, R2, and the return wire.

Worked example 6

Choose the source − terminal as 0 V. Moving to source + raises the voltage to 12 V. Then R1 and R2 bring it back down to 0 V. With a clockwise current I, the source change is +12 V and each resistor change is −IR:

+12 − I(1000) − I(2000) = 0

12 = 3000I, so **I = 4 mA**. Substitute: +12 − 4 − 8 = 0 V. This is the same series answer, now obtained directly from KVL.

Traversal	Voltage change
Source from − to +	+Vs, a rise
Source from + to −	−Vs, a drop
Resistor in the chosen current direction	−IR, a drop
Resistor opposite the chosen current direction	+IR, a rise

A second source example

Suppose a loop contains a 12 V source, an opposing 3 V source, and 3 kΩ total resistance. In your chosen direction: +12 − 3 − I(3 kΩ) = 0. Then **I = 3 mA**. If the sources aid each other, the rises add and the current is 15/3 = **5 mA**. Always inspect source polarity first.

Interview distinction: KCL deals with currents at a node; KVL deals with voltage changes around a loop. Ohm’s Law links voltage and current for a resistor. Use all three together.

Voltage Dividers and Loading

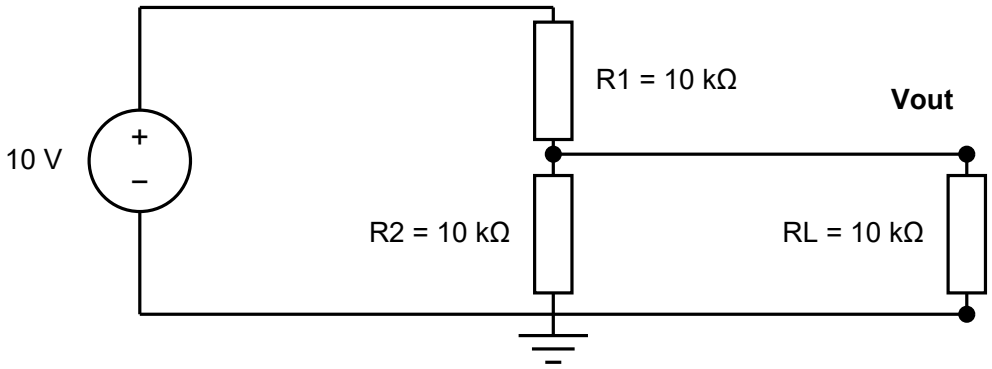


Figure 7. The load RL connects from Vout to 0 V, so RL is parallel with R2.

No load Vout = Vin × R2/(R1 + R2)

Vout means the voltage at the middle connection, compared with the bottom 0 V wire. With no RL connected, R1 and R2 are equal and carry the same current. They split the 10 V equally: 5 V across each. Therefore Vout = 10 − 5 = **5 V**. The formula above calculates that same result.

Worked example 7

1. Add RL = 10 kΩ. Replace R2 and RL with Rb = R2 || RL = **5 kΩ**. The symbol || means parallel.

2. Now Vout = 10 × 5/(10 + 5) = **3.333 V**.

3. Source current = 10/15 kΩ = **0.667 mA**. Current in R2 = 3.333/10 = 0.333 mA. Load current is also 0.333 mA.

4. Check: the currents leaving Vout add to 0.667 mA. The load lowered the effective bottom resistance and pulled Vout down.

Why does the voltage fall? The added load creates another path for current. More current now flows through R1, so R1 drops more voltage. Less of the original 10 V is left at Vout.

What if the load is a voltmeter

A real voltmeter is also a load. Its input resistance appears in parallel with R2. If that resistance is much larger than R2, the effect is small. If it is comparable, the reading can be significantly lower than the original unloaded voltage. Page 12 calculates this.

When the divider formula is valid

• Identify the resistor from Vin to Vout and the equivalent resistance from Vout to the reference.

• Include every relevant output load before using the formula.

• A basic resistor divider is useful for signal scaling or biasing; it does not hold a fixed output when load current changes.

Try it: Vin = 12 V, upper R1 = 2 kΩ, lower R2 = 1 kΩ, no load. **Answer:** Vout = 4 V, I = 4 mA.

Resistors in Real Circuits

A resistor can limit LED current, divide a voltage, or help keep an input at a known voltage when nothing else drives it. You can also measure current by measuring the voltage across a known resistor. Check its resistance, tolerance, and power rating.

Tolerance and power

Worked example 8: A 1 kΩ resistor with ±5% tolerance may be 950 to 1050 Ω. Across an ideal 5 V supply, current is highest at 950 Ω: 5/950 = **5.263 mA**. It is lowest at 1050 Ω: **4.762 mA**.

For 12 V across 1 kΩ, P = 12²/1000 = **0.144 W**. A 0.125 W resistor is below this dissipation. A 0.25 W part may be appropriate if its temperature and derating limits allow it. Avoid choosing a rating exactly equal to the calculated dissipation.

Reading common markings

For a four band resistor, the first two bands are digits, the third is the multiplier, and the fourth is tolerance. Example: **brown black red gold** = 10 × 100 Ω = **1 kΩ ±5%**. Gold tolerance is ±5%; silver is ±10%.

Color	Digit	Multiplier
Black	0	×1
Brown	1	×10
Red	2	×100
Orange	3	×1000
Yellow	4	×10000
Green	5	×100000
Blue	6	×1000000
Violet	7	×10000000
Gray	8	×100000000
White	9	×1000000000

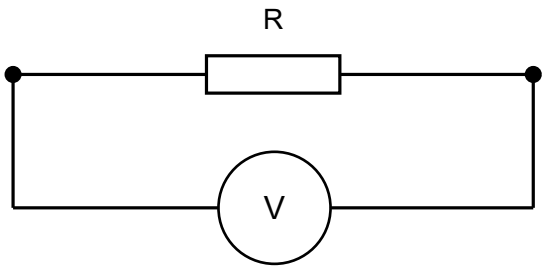
Common three digit SMD code: first two digits then a power of ten. **103** = 10 × 10^3 = **10 kΩ**; **472** = 47 × 10^2 = **4.7 kΩ**. Other marking systems exist; confirm the component documentation.

Fault clues: An open resistor can stop branch current. A shorted resistor can remove a voltage drop and increase current elsewhere. Measure resistance with power off; an in-circuit reading includes other paths.

Voltmeter Ammeter and Ohmmeter

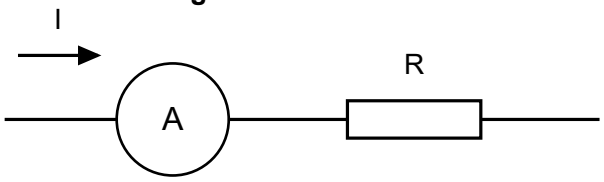
How to connect and why internal resistance matters

Voltage across R



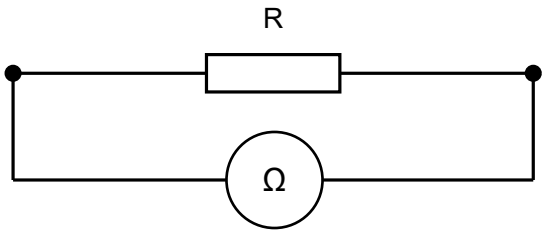
Red probe (+) Black probe (COM)

Current through R



Open the path and insert the meter

Resistance across an isolated resistor



Power off

Capacitors discharged

Isolate R if other paths exist

Figure 8. The meter connection changes with what you measure. The resistor remains in the circuit for voltage measurement.

Mode	Connection	Ideal behavior	Real behavior
Voltmeter	Across two nodes in parallel	Infinite input resistance; draws no current	Finite input resistance draws some current and loads the circuit.
Ammeter	In the current path in series	Zero resistance; no added drop	Shunt, leads, and fuse add a voltage drop called burden voltage.
Ohmmeter	Across the unpowered component	Applies a known stimulus to infer R	Internal source and range circuitry; no single universal internal resistance.

Voltage measurement

Black lead to **COM**, red to **VΩ**. Select DC V for DC circuits. Connect across the desired nodes: red at the point expected to be higher, black at the reference. A negative display means the red point is at a lower potential. Start with a suitable range and check the instrument's ratings.

Current measurement

Turn the circuit power off before opening the path. Put black in COM and red in the appropriate fused **mA or A** jack. Choose DC current and a range that covers the expected current. Insert the meter in series, then power the circuit and read. Power off before removing it; return the red lead to VΩ afterward.

Critical mistake: An ammeter connected directly across a voltage source creates a low resistance path. Large current can damage the meter or circuit or blow the fuse. Do not use that connection to measure source current.

Meter Loading and Resistance Tests

Worked example 9 on voltmeter loading

A 10 V divider has upper and lower resistors of 1 MΩ each. Without a meter, Vout = 5 V. A meter with 10 MΩ input resistance connects across the lower resistor. The lower resistance becomes 1 MΩ || 10 MΩ = **0.9091 MΩ**.

Vread = 10 × 0.9091/(1 + 0.9091) = 4.7619 V

What happened? The meter is an extra current path, like RL on page 9. It changes the circuit slightly, so it reads 4.7619 V instead of 5 V. That is **4.76% lower**. A 10 MΩ input resistance is an example; check the actual meter specification. [3]

Optional deeper formula: Vread = Vth × Rv/(Rth + Rv). Vth is the voltage before connecting the meter, Rth is the equivalent circuit resistance at the measuring points, and Rv is meter input resistance. Here Rth = 1 MΩ || 1 MΩ = 0.5 MΩ.

Worked example 10 on ammeter burden

A 5 V source and 100 Ω resistor normally give 50 mA. Suppose the meter adds an effective 2 Ω in series. Imeasured = 5/(100 + 2) = **49.02 mA**. Meter drop = I × 2 = **98.04 mV**. The meter changed the current by about **1.96%**. Real burden depends on the meter and selected range. [3]

How an ohmmeter works

It supplies a small test current or voltage and measures the response to infer R = V/I. A traditional analog ohmmeter contains a battery and range resistors; a digital meter often uses controlled test current. Its internal resistance is not simply “zero” or “infinite.” Use the model and manual for the selected range.

1. Disconnect circuit power. Discharge capacitors through an appropriate resistor and verify the voltage is near zero.

2. Use COM and VΩ, select Ω, and place probes across the part. Isolate one terminal if parallel circuit paths could affect the reading.

3. For low resistance, account for lead and contact resistance. A four wire measurement uses separate current and voltage leads to reduce this error. [7]

4. Read the unit prefix. OL usually means above range or an open path in Ω mode; it does not prove a component is faulty without context.

Example: Two 10 kΩ resistors are in parallel. A powered-off in-circuit test across either reads about **5 kΩ**, not 10 kΩ. Isolating that resistor removes the alternate path. A continuity beep indicates a path below the meter’s threshold, not necessarily zero ohms. [4]

Internal resistance of a real source

Think of a real source as a perfect source Vs with a small resistor r inside it. The current passes through r and the external load RL: I = Vs/(r + RL). For Vs = 5 V, r = 10 Ω, and RL = 90 Ω, I = **50 mA**. The inside resistor drops 0.05 × 10 = **0.5 V**, leaving **5 − 0.5 = 4.5 V** at the load. This is one simple source model.

Capacitors and Their Connections

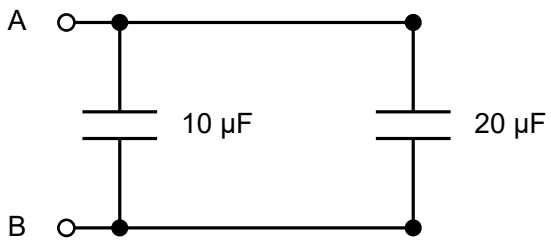
A capacitor stores electrical energy by building up opposite charges on two conductors separated by an insulator. **Capacitance tells you how much charge it holds for each volt.** At 1 V, a 10 µF capacitor holds 10 µC of charge magnitude; a 20 µF capacitor holds 20 µC.

$Q = CV$ Stored energy = $\frac{1}{2}CV^2$

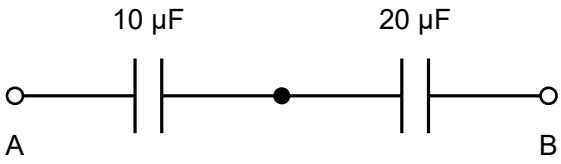
Q means charge, C means capacitance, and V means the voltage between the two ends. More capacitance needs more charge for the same voltage. For current, $i = C\,dv/dt$ means **faster voltage change causes more current**. If voltage stays constant, ideal capacitor current is zero. Real capacitors have small leakage and losses.

Connection	Equivalent capacitance	What is shared
Parallel	Ctotal = C1 + C2 + …	Same voltage; capacitances add.
Series	1/Ctotal = 1/C1 + 1/C2 + …	Same charge magnitude under the usual initially uncharged series assumptions.

Parallel capacitors



Series capacitors



Middle point with no other connection

Figure 9. In parallel both capacitors share A and B. In series the junction between them is floating with no other connection.

Worked example 11

10 µF and 20 µF in parallel give **30 µF**. In series they give (10 × 20)/(10 + 20) = **6.667 µF**. For the series pair across 12 V, assume both started uncharged and their middle point has no other connection. Q = 6.667 µF × 12 V = **80 µC**.

V across 10 µF = 80 µC / 10 µF = **8 V**. V across 20 µF = **4 V**. The smaller capacitor gets more voltage for the same charge. In real series capacitors, leakage and initial conditions affect voltage sharing; ratings cannot simply be assumed to divide equally.

Initial state versus final state

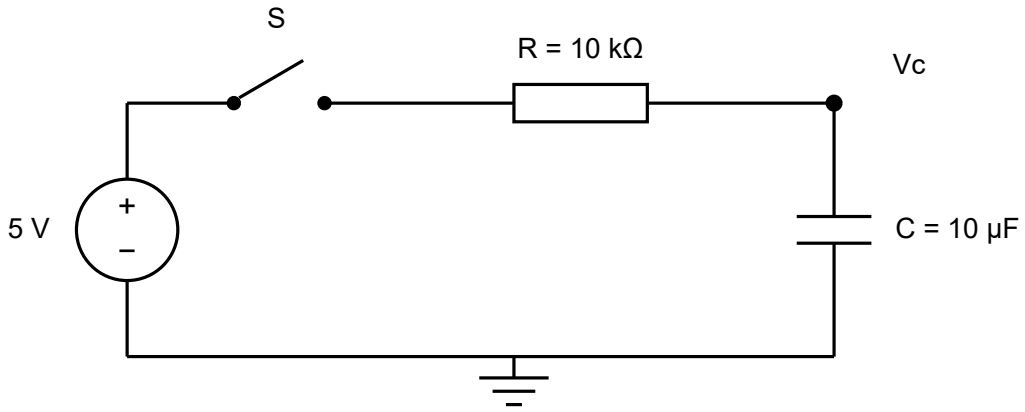
• In these RC circuits, the capacitor voltage does not suddenly jump when the switch closes. It starts at the voltage it already had and then changes.

• If it starts at 0 V, it has 0 V across it at the first instant. That is why people say “initially like a short.” A capacitor that already has charge does not start at 0 V.

• After a long time with steady DC, ideal capacitor current becomes zero. That is why people say “like an open circuit.” It can still hold voltage.

Practical checks: Observe polarity on electrolytic capacitors, select a suitable voltage rating, and check the datasheet. A shorted capacitor can pull down a supply; an open or high ESR capacitor can worsen filtering. A normal capacitance reading alone does not prove ESR or leakage is good.

RC Charging and Discharging



Close S at t = 0. Initially Vc = 0 V.

Figure 10. The initially uncharged capacitor charges through R after switch S closes.

τ = RC Vc(t) = Vs[1 − exp(−t/τ)] for charging from 0 V

τ is pronounced tau and means time constant. R = 10 kΩ and C = 10 μF give τ = 10000 × 0.000010 = **0.1 s**. At first Vc = 0 V, so R has all 5 V across it: I = 5/10 kΩ = **0.5 mA**. As Vc rises, less voltage is left across R and the current falls.

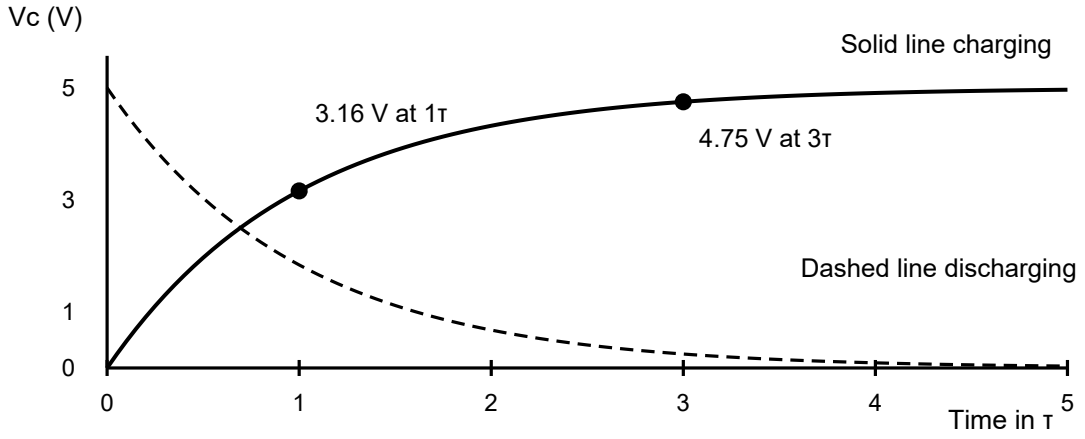


Figure 11. Charging toward 5 V and discharging from 5 V, each with the same RC time constant.

Worked example 12: At 0.1 s = 1τ, Vc = 5[1 − exp(−1)] = **3.16 V**. Current = (5 − 3.16)/10 kΩ = **0.184 mA**. At 0.3 s = 3τ, Vc = **4.75 V**. At 0.5 s = 5τ, Vc = **4.97 V**, about 99.3% of final value. [5]

Discharging through R Vc(t) = Vinitial exp(−t/RC)

For discharge from 5 V through 10 kΩ, after 0.1 s the voltage is **1.84 V**. The source must be disconnected from the charging path and a discharge path provided. The general first order expression is Vc(t) = Vfinal + (Vinitial − Vfinal)exp(−t/τ).

How to read the formula: exp(x) means e raised to x. At t = τ, t/τ = 1 and exp(−1) ≈ 0.368. Charging therefore reaches 1 − 0.368 = 0.632, or 63.2% of the final voltage. In this simple circuit, the R in τ = RC is 10 kΩ.

Extra AC concept: |Xc| = 1/(2πfC). As frequency increases, a capacitor offers less opposition to a changing signal. A simple RC low pass measured across C has cutoff fc = 1/(2πRC). The glossary explains these terms.

Diodes and Their Models

Think of a diode as a component that lets current flow much more easily in one direction. Its ends are the **anode A** and **cathode K**. Forward current goes from A to K. The line in the symbol identifies K; many physical diodes also mark K with a band.

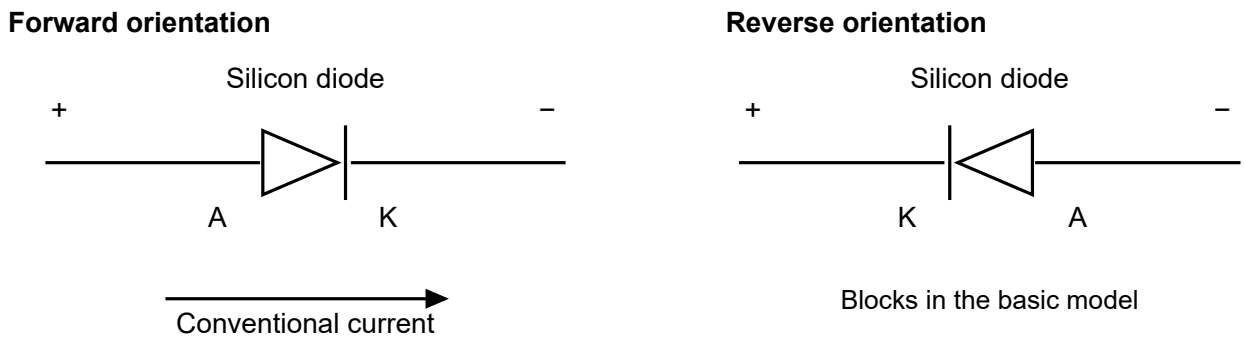


Figure 12. The bar identifies the cathode. Here the left terminal is positive relative to the right; that forward biases the first diode and reverse biases the second.

Model	Forward behavior	Reverse behavior
Ideal diode	ON with 0 V drop when carrying forward current	OFF with 0 A until an excluded breakdown condition.
Constant drop model	ON with a specified drop such as 0.7 V for a silicon diode	OFF with 0 A if reverse breakdown is excluded.
Real diode	Nonlinear I-V curve; drop depends on current and temperature	Leakage occurs; breakdown can occur at sufficient reverse voltage.

ON means conducting; OFF means blocking in the chosen model. A model is a simpler rule used for a calculation. The 0.7 V rule is approximate: a real diode's forward drop changes with type, current, and temperature. Use the given rule in a problem and the datasheet for an actual part.

Common uses

• Rectifier: conducts on the appropriate polarity of AC. A half wave rectifier uses one diode; a bridge has two conducting diode drops in each current path.

• LED: emits light under forward current and needs current limiting.

• Zener: designed for specified reverse breakdown operation with limited current. Ordinary rectifier breakdown is not a normal design assumption.

Testing a diode with a digital multimeter

Turn power off, discharge capacitors, and isolate the diode if other paths interfere. Select diode mode with leads in COM and VΩ. Red on A and black on K should show a forward voltage for a conducting diode; reverse the leads and expect OL for a normal blocking diode within the meter's test limits. [6]

Near zero in both directions suggests a short. OL in both directions suggests an open or insufficient meter test voltage, for example for some LEDs or diode stacks. Confirm the part and meter limits before declaring a fault. Diode mode does not normally measure Zener breakdown voltage. [6]

Solving Diode and LED Circuits

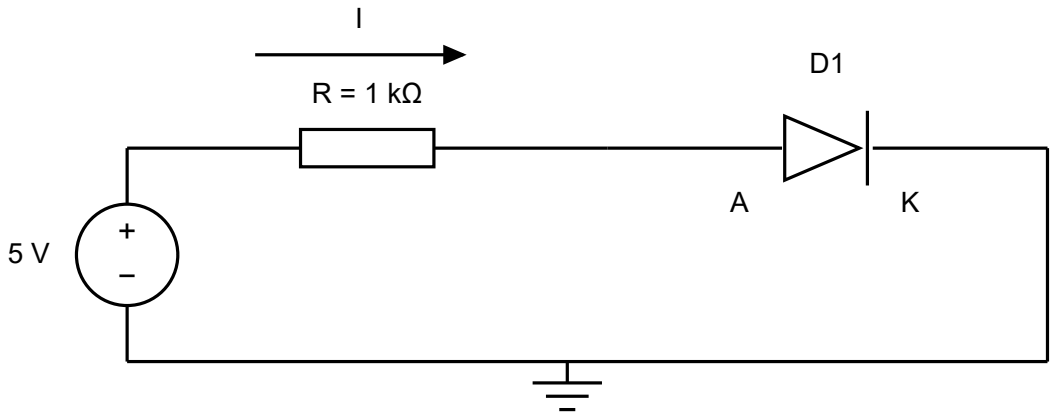


Figure 13. A 5 V source drives a forward oriented diode through 1 kΩ.

Worked example 13

1. State the model: silicon diode with a constant forward drop of **0.7 V**, and reverse current zero below breakdown.

2. Assume the diode is ON. KVL: 5 − IR − 0.7 = 0.

3. I = (5 − 0.7)/1 kΩ = **4.3 mA**. This current flows A to K, consistent with the ON assumption.

4. Diode voltage = 0.7 V; resistor voltage = 4.3 V. PR = I^2R = **18.49 mW**.

Reverse the diode: In the stated model it is OFF, so I = 0 A and the resistor drop is 0 V. The cathode is then at 5 V and anode at 0 V: VAK = −5 V. The diode blocks the source voltage; check the real part's reverse rating.

Worked example 14 on an LED

A 5 V source must drive an LED at 10 mA. Assume VF = 2.0 V at the target current. The resistor must drop 5 − 2 = 3 V.

R = (Vs − VF)/I = 3/0.010 = 300 Ω

A common 330 Ω choice gives I ≈ 3/330 = **9.09 mA**. Resistor power ≈ 3^2/330 = **27.3 mW**. Check the LED's current limit, resistor tolerance, maximum supply voltage, minimum VF, and resistor power rating. [1]

How to handle an unfamiliar diode problem

1. Mark A and K. Choose ideal or stated forward drop model.

2. Assume ON or OFF, replace the diode with that model, and solve the resistor circuit.

3. Check consistency: an ON result must allow forward current; an OFF result must not demand conduction under your chosen model.

4. If the assumption fails, try the other state. For several diodes, check each state after solving.

Diodes in series and parallel

Series: The same current flows through each diode and forward drops add. Two forward silicon diodes with 0.7 V each, a 5 V source, and 1 kΩ in series give I = (5 − 1.4)/1 kΩ = **3.6 mA**.

Parallel: Diodes share voltage, but mismatched forward characteristics can cause unequal current sharing. For parallel LED branches, use current limiting for each branch. A resistor before or after one LED in the same unbranched series path has the same current limiting effect.

Troubleshooting with Circuit Analysis

Start by predicting normal voltages and currents. Compare measurements with that prediction, then choose a test that distinguishes possible faults. One unusual reading is a clue, not a complete diagnosis.

Scenario A a divider gives the wrong voltage

Use a 10 V source with upper R1 = 10 kΩ and lower R2 = 10 kΩ. Measure Vout relative to 0 V with a high input resistance meter. The normal unloaded output is about 5 V. Assume one ideal fault at a time.

Vout	Possible single fault	Why
Near 0 V	R1 open or R2 short	The output is pulled to the bottom node.
Near 10 V	R1 short or R2 open	The output is connected toward the supply with no normal divider action.
Between 0 V and 5 V	Extra load or incorrect resistor value	The bottom effective resistance may have decreased.
Unstable or unexpected	Loose reference, intermittent connection, or other fault	Check actual connections and supply before inferring a component fault.

Next tests: confirm Vin and the reference with voltage mode. Power off and discharge capacitors before checking resistor values, continuity, or solder joints. Isolate a component if parallel paths affect the result.

Scenario B an LED is dark

1. Confirm supply voltage and LED polarity. Check whether the circuit is actually closed.

2. Measure the resistor drop in voltage mode. Use I = VR/R to infer the branch current without inserting an ammeter.

3. If VR ≈ 0 V, there may be no current: consider an open wire, open LED, or reverse LED. If VR ≈ Vs, consider a shorted LED, but check connections.

4. Power off before diode testing or resistance checks. Compare measured forward voltage with the actual LED type and operating current.

What to say when measurements disagree

“I’ll verify my reference, meter mode, lead jack, and connections. Then I’ll compare the measured node voltages with the circuit calculation. I’ll choose a power-off component test to separate likely causes and check whether the instrument itself is loading the circuit.”

Capacitor clue: A slowly rising Ω reading may be the meter charging a capacitor. It is not a dependable standalone test of capacitance, leakage, or ESR.

Technical Questions on Circuit Analysis

Answer aloud first then compare with the model answer

1. What is the difference between voltage and current?
Voltage is potential difference between two points; current is charge flow per second through a path. Measure voltage across and current through.

2. State Ohm's Law and its limitation.
V = IR for an ohmic resistor with the relevant R. The V is across that resistor. A diode does not have one constant R over its whole operating range.

3. Why is current the same in series?
There is no branching path between the series elements. KCL requires the current entering their shared node to equal the current leaving it.

4. Why is voltage the same in parallel?
Every branch connects to the same two nodes, so the difference between those node voltages is identical.

5. What happens when another parallel resistor is added?
Equivalent resistance decreases. For a fixed ideal voltage supply, total current increases. Existing resistor branch currents stay the same if their voltage stays fixed.

6. How do you recognize series and parallel on a messy diagram?
Trace and label nodes. Series elements share an unbranched junction; parallel elements share both nodes. The visual position of symbols does not decide it.

7. State KCL and KVL.
KCL: algebraic current sum at a node is zero. KVL: algebraic voltage changes around a closed loop sum to zero in the circuit model.

8. What does a negative calculated current mean?
The actual current flows opposite to my assumed arrow. I retain the sign while checking KCL and interpreting the result.

9. Can an open circuit have voltage across it?
Yes. It has no current through the open path, but its two sides can be at different potentials. A battery has voltage even with no load connected.

10. When should you use nodal analysis?
When unknown node voltages are convenient or the network cannot be reduced easily. Pick a reference and write each branch current as a voltage difference divided by its resistance.

11. Why did a voltage divider output drop when a load was attached?
The load is parallel with the lower resistor. It lowers the bottom effective resistance and changes the divider ratio.

12. A resistor's voltage doubles. What happens to power?
With fixed resistance, $P = V^2/R$, so power becomes four times larger. Check the resistor power rating and operating temperature.

Follow-up drill: For Figure 4, explain why R1 does not carry the same current as R2. Then calculate all three currents from scratch.

Technical Questions on Components

Meters capacitors and diodes

13. Why does a voltmeter need high input resistance?
To draw little current and reduce loading. Ideally its resistance is infinite; a real meter has finite input resistance or impedance.

14. Why does an ammeter need low resistance?
It is inserted in series, so low resistance reduces its added voltage drop and its effect on the circuit current. The actual burden depends on range and meter design.

15. Why must resistance be measured with power off?
The meter provides its own test stimulus. External voltage can invalidate the result or damage the meter. Capacitors must be discharged too.

16. Why might a 10 kΩ resistor measure 5 kΩ on a board?
Another 10 kΩ path may be in parallel. Isolate one terminal and remeasure before concluding that the resistor is wrong.

17. Does a continuity beep prove a good component?
No. It proves only that the measured path meets the meter's continuity threshold. Other parallel paths can create the beep.

18. What is the internal resistance of an ohmmeter?
It depends on the meter and range. The meter has an internal test source and measurement circuitry, so it is not described by a universal zero or infinite resistance.

19. What happens to a capacitor just after switching and after a long time?
Its voltage initially retains its pre-switch value. In ideal DC steady state its current is zero, so it is an open circuit. "Initially a short" applies only when initially uncharged.

20. What does one RC time constant mean?
For charging from zero, the capacitor reaches about 63.2% of final voltage. During discharge it falls to about 36.8% of its initial voltage.

21. How do you combine capacitors?
Parallel capacitances add. For series capacitors, reciprocal capacitances add. These are the opposite patterns to resistor combination formulas.

22. Is a silicon diode always exactly 0.7 V?
No. 0.7 V is a useful constant drop approximation. Actual forward voltage depends on current, temperature, and the diode type.

23. Why does an LED need a current limiting resistor?
Its nonlinear characteristic can allow excessive current from a voltage supply. Use R = (Vs − VF)/I and check worst-case current and component ratings.

24. How would you test a diode?
Power off, discharge capacitors, and isolate if needed. In diode mode compare both polarities: a normal diode conducts forward and blocks reverse within the meter's test limits.

Practice Problems Part One

Try without the solutions and show your equations

Allow about 20 minutes for this page. State assumptions, show units, and check the answer. Ideal wires and sources are assumed unless the problem says otherwise. Keep three significant digits where appropriate.

Problem 1 Ohms Law and power

A 330 Ω resistor is directly across 3.3 V. Find current and resistor power. Would a 0.125 W rating exceed the calculated dissipation?

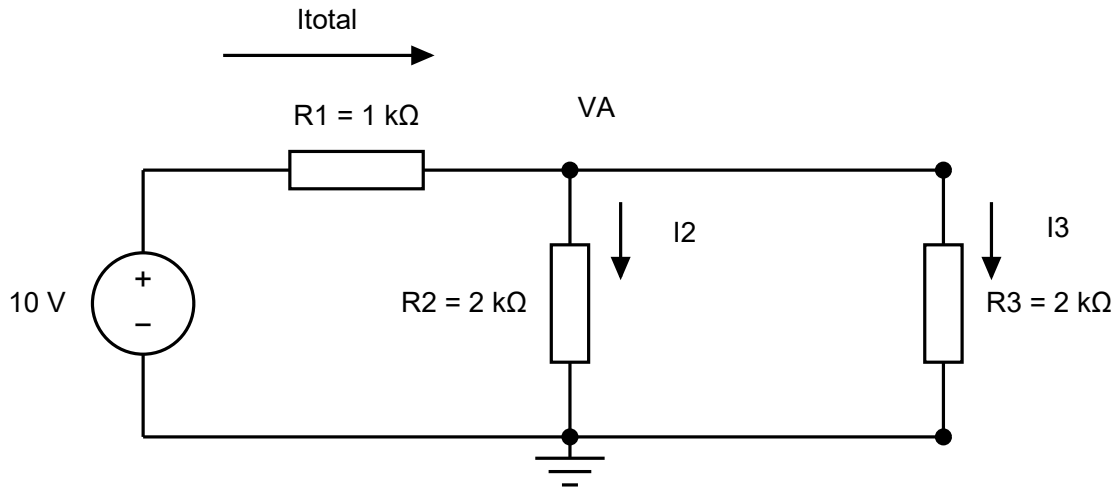
Problem 2 series circuit

A 9 V source supplies 1 kΩ and 2 kΩ in series. Find total resistance, current, and the voltage across each resistor. Write one KVL check.

Problem 3 parallel circuit

2 kΩ and 3 kΩ are in parallel across 6 V. Find equivalent resistance, each branch current, and source current. Check KCL.

Problem 4 mixed circuit



Practice circuit A. Find total resistance, source current, VA, I2, and I3.

Problem 5 one unknown node

Vx connects through 2 kΩ to a known 12 V node, through 4 kΩ to a known 0 V node, and through 4 kΩ to a known 4 V node. All known voltages share the same reference. Find Vx and each branch current using KCL. Use currents leaving Vx as positive.

Problem 6 loaded divider

Vin = 12 V. Upper R1 = 2 kΩ; lower R2 = 2 kΩ. Find unloaded Vout. Then attach RL = 2 kΩ from Vout to 0 V and find the loaded output, source current, and load current.

Do not skip the connection check: In Problem 4, which two resistors share the same two nodes? In Problem 6, which resistance must change before using the divider formula?

Practice Problems Part Two

Meters capacitors diodes and troubleshooting

Problem 7 voltmeter loading

A 10 V divider has upper and lower resistors of 100 kΩ each. A voltmeter with 1 MΩ input resistance measures the lower resistor voltage. Find the reading and its percentage error relative to the unloaded output.

Problem 8 ammeter burden

A 3 V source drives a 100 Ω resistor. An ammeter adds an effective 5 Ω in series. Find original current, measured current, and meter voltage drop.

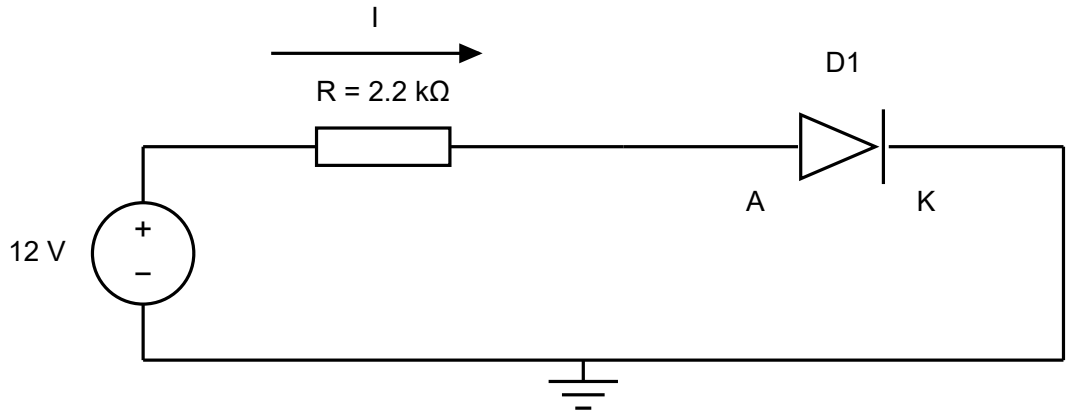
Problem 9 capacitor combinations

6 µF and 3 µF capacitors are first connected in parallel and then in series. Find each equivalent. For the ideal series pair across 9 V, assume zero initial charge at the floating junction. Find the charge magnitude and voltage across each capacitor.

Problem 10 RC timing

A 10 µF capacitor charges from 0 V toward 5 V through 10 kΩ. Find τ, the initial current, and capacitor voltage at t = τ and t = 5τ. Separately, if it starts at 5 V and discharges through 10 kΩ, find the voltage after τ.

Problem 11 diode circuit



Practice circuit B. Use a constant forward drop of 0.7 V.

Find diode current, resistor voltage, and resistor power. Then reverse the diode, assume breakdown does not occur, and find current and VAK.

Problem 12 troubleshooting and meter choice

An unloaded 10 V divider uses upper and lower 10 kΩ resistors. A high input resistance voltmeter reads Vout ≈ 0 V. Name two possible single component faults. Explain what you would measure next and when you must turn power off.

Extra challenge: You want to determine current through a known resistor without breaking the circuit. Which meter mode and formula could you use, and what assumptions affect accuracy?

Practice Solutions Part One

Problem 1

I = 3.3/330 = **0.010 A = 10 mA**. P = 3.3 × 0.010 = **0.033 W = 33 mW**. A 0.125 W rating exceeds the calculated dissipation. Actual selection still depends on the resistor's allowed power at the operating temperature.

Problem 2

Rt = 1 + 2 = **3 kΩ**. I = 9/3 = **3 mA**. V1 = 3 × 1 = **3 V**; V2 = 3 × 2 = **6 V**. KVL: +9 − 3 − 6 = 0 V. Both resistors carry 3 mA.

Problem 3

Rp = (2 × 3)/(2 + 3) = **1.2 kΩ**. I1 = 6/2 = **3 mA**; I2 = 6/3 = **2 mA**. Itotal = **5 mA**. Check: 6/1.2 = 5 mA and 3 + 2 = 5 mA.

Problem 4

Rp = 2 kΩ || 2 kΩ = **1 kΩ**. Rt = 1 + 1 = **2 kΩ**. Itotal = 10/2 = **5 mA**. R1 drops 5 mA × 1 kΩ = **5 V**, so VA = **5 V**. I2 = I3 = 5/2 = **2.5 mA**. Checks: 2.5 + 2.5 = 5 mA and 10 − 5 − 5 = 0 V.

Problem 5

(Vx − 12)/2 + Vx/4 + (Vx − 4)/4 = 0

Multiply by 4: 2Vx − 24 + Vx + Vx − 4 = 0. Hence 4Vx = 28 and **Vx = 7 V**. Current from Vx toward 12 V: (7 − 12)/2 = **−2.5 mA**, meaning 2.5 mA enters from the 12 V source. Toward ground: 7/4 = **1.75 mA**. Toward 4 V: (7 − 4)/4 = **0.75 mA**. KCL: −2.5 + 1.75 + 0.75 = 0 mA.

Problem 6

Unloaded: 12 × 2/(2 + 2) = **6 V**. Loaded bottom resistance: 2 || 2 = **1 kΩ**. Vout = 12 × 1/(2 + 1) = **4 V**. Source current = 12/3 = **4 mA**. Load current = 4/2 = **2 mA**; lower resistor current is also 2 mA. The output dropped because the load changed the bottom equivalent resistance.

If you missed Problems 4 to 6: Redraw the nodes before retrying. A correct formula cannot repair an incorrect connection assumption.

Practice Solutions Part Two

Problem 7

Bottom equivalent = 100 kΩ || 1000 kΩ = **90.909 kΩ**. Vread = 10 × 90.909/(100 + 90.909) = **4.7619 V**. Unloaded value = 5 V. Signed percentage error = (4.7619 − 5)/5 × 100 = **−4.7619%**. The meter loads the divider.

Problem 8

Original I = 3/100 = **30 mA**. With meter: I = 3/105 = **28.571 mA**. Meter drop = 0.028571 × 5 = **0.142857 V**. The reading is the current in the modified circuit, not the undisturbed circuit.

Problem 9

Parallel C = 6 + 3 = **9 µF**. Series C = (6 × 3)/(6 + 3) = **2 µF**. Series charge Q = 2 µF × 9 V = **18 µC**. V6µF = 18/6 = **3 V**; V3µF = 18/3 = **6 V**. Check: 3 + 6 = 9 V.

Problem 10

τ = 10000 × 0.000010 = **0.1 s**. Initial I = 5/10000 = **0.5 mA**. At τ, Vc = 5[1 − exp(−1)] = **3.1606 V**. At 5τ, Vc = 5[1 − exp(−5)] = **4.9663 V**. During discharge, Vc(τ) = 5exp(−1) = **1.8394 V**.

Problem 11

ON model: I = (12 − 0.7)/2.2 kΩ = **5.136 mA**. VR = **11.3 V**. PR = 11.3²/2200 = **0.05804 W = 58.04 mW**. Reversed and below breakdown: I = **0 A**, VR = 0 V, and VAK = **−12 V**.

Problem 12

Possible single faults are **upper resistor open** or **lower resistor short**. Confirm the 10 V source and ground reference in voltage mode. Then power off and discharge capacitors before measuring resistance or continuity. Isolate resistor terminals if parallel paths affect the test. Do not claim one definite fault from Vout alone.

Extra challenge

Measure the voltage across the resistor in **voltage mode**, then infer I = VR/R. Account for resistor tolerance, meter accuracy and loading, temperature, and the resistor's actual value. This can avoid opening a delicate signal path.

Personal study target: Aim to solve at least 10 of 12 problems independently, explain the corrections to every missed problem, and make no unsafe meter connection errors. This is a practice target, not an ADI hiring score.

Final Recall and Technical Rehearsal

Topic	What to remember
Ohm and power	V = IR; P = VI = I²R = V²/R for a resistor.
Resistors	Series: sum R and same I. Parallel: sum 1/R and same V.
Kirchhoff	KCL: current in = current out. KVL: voltage changes around loop sum to zero.
Divider	Vout = Vin × Rbottom/(Rtop + Rbottom); include the output load.
Meters	V across; A in series; Ω with power off and capacitors discharged.
Internal resistance	Voltmeter high; ammeter low; ohmmeter source and range dependent.
Capacitors	Parallel: sum C. Series: sum 1/C. Voltage cannot jump without impulse current.
RC	τ = RC; charge 63.2% at τ and 99.3% at 5τ; discharge 36.8% at τ.
Diodes	Check A and K, state the model, assume ON or OFF, and verify consistency.

A fifteen minute technical rehearsal

Spend 5 minutes solving Figure 4 while speaking aloud. Spend 4 minutes explaining KCL, KVL, and the meter connections. Spend 3 minutes explaining RC charging and an LED resistor calculation. Spend 3 minutes diagnosing the divider fault on page 17. If you get stuck, state what is known, choose a reference, and write the governing law before guessing.

Reference reading

The worked values, practice circuits, and model answers in this reviewer are original teaching examples. Use these primary sources for further study and instrument-specific details. Links are clickable. Accessed October 14 2026.

[1] ADI Ohms Law experiment

[2] ADI review of Kirchhoffs circuit laws

[3] Fluke 8050A manual on input resistance and burden voltage

[4] Fluke measuring resistance and parallel paths

[5] ADI transient response of an RC circuit

[6] Fluke testing diodes with a digital multimeter

[7] Fluke 8845A and 8846A manual on four wire resistance

YouTube Guide for Your Study Plan

Watch in this order and solve a matching problem after each topic

Start with the first three links because circuit analysis is your priority. Video titles below are clickable; the short URLs also work from a printed copy. Pause before a solution is shown and try the calculation on paper. These are supporting lessons from independent educators.

1 Ohms Law

Ohm’s Law
The Organic Chemistry Tutor | youtu.be/_rSHqvjDksg

Pages 2 and 3; Problem 1. Focus on voltage across a resistor, current through it, and units.

2 Series and parallel circuits

Resistors In Series and Parallel Circuits - Keeping It Simple!
The Organic Chemistry Tutor | youtu.be/dEuBykX5_B4

Pages 4 to 6; Problems 2 to 4. Trace the nodes, reduce groups, and recover branch values.

3 KCL and KVL

Kirchhoff’s Law, Junction and Loop Rule, Ohm’s Law
The Organic Chemistry Tutor | youtu.be/2Zu3ppq3n8I

Pages 7 and 8; Problem 5. Focus on current direction and consistent voltage rise and drop signs.

4 Using a multimeter

THE BEST Multimeter tutorial (HD)
Afrotechmods | youtu.be/bF3OyQ3HwfU

Pages 11 and 12; Problems 7, 8, and 12. Explain the connection, mode, jack, and effect on the circuit.

5 Capacitor basics

Capacitors - Basic Introduction - Physics
The Organic Chemistry Tutor | youtu.be/BimpNou0orc

Page 13; Problem 9. Understand Q = CV and compare resistor and capacitor connection formulas.

6 RC charging and discharging

RC Circuits Physics Problems and Time Constant Explained
The Organic Chemistry Tutor | youtu.be/PLQrPqYlPmI

Page 14; Problem 10. Distinguish initial and final behavior, and calculate one time constant.

7 Diodes and rectification

Diode Tutorial and How to build an AC to DC power supply
Afrotechmods | youtu.be/cyhzpFqXwdA

Pages 15 and 16; Problem 11. Focus on A and K, forward drop, and why a bridge has two conducting diodes.

Before the interview: Return to the paper problems. Watching a clear explanation helps most when you can reproduce the method afterward without the video.

DC and AC Explained

Understanding direction polarity and changing signals

DC means direct current

DC current flows in one direction. A DC voltage source keeps one polarity: its positive terminal stays positive relative to its negative terminal. A battery is a familiar example. **DC does not have to be perfectly constant.** It can vary or contain ripple while retaining its polarity; steady DC is the constant case used in most worked examples here.

AC means alternating current

AC current changes direction over time. An AC voltage changes polarity, so the terminal that is positive during one part of a cycle becomes negative during another part. A sine wave is a common AC waveform, but AC can have other shapes. Direction changes do not mean the circuit is broken.

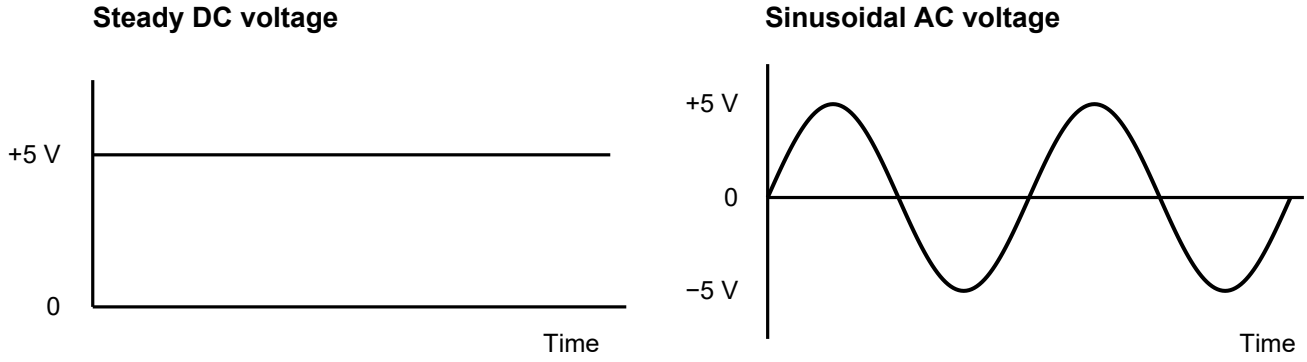


Figure 14. Example voltages: steady DC stays at +5 V. The AC sine wave crosses 0 V and varies between +5 V and −5 V. Negative voltage means reversed polarity relative to the chosen reference.

Feature	DC	AC
Direction or polarity	One direction or one polarity	Alternates over time
Typical example	Battery supply	Sine wave from a signal generator
Meter setting	DC V or DC A	AC V or AC A
Capacitor behavior	Ideal current is zero after DC settles	Changing voltage can produce capacitor current

How to describe an AC signal

Frequency f is cycles per second, measured in hertz (Hz). **Period T** is the time for one cycle: T = 1/f. For a 50 Hz example, T = 0.02 s = 20 ms. **Peak voltage** is the largest magnitude measured from zero. In the diagram, Vpeak = 5 V and peak-to-peak voltage is 10 V.

RMS voltage tells you the equivalent DC voltage that produces the same average heating in a resistor. For a pure sine wave with no DC offset, Vrms = Vpeak/√2. The illustrated 5 V peak sine wave has **3.54 V RMS**. Its average over a complete cycle is zero, but it still delivers power to a resistor.

Meter and analysis connection: Choose the setting for the quantity you want; AC mode and DC mode can show different parts of a mixed signal. Check the meter manual for bandwidth and RMS behavior. Ohm's Law applies instant by instant to an ideal resistor. For a capacitor, use i = C dv/dt or an appropriate AC impedance model.

Definition of Terms

Circuit quantities laws and connections

Circuit analysis Finding voltages, currents, and power from a circuit's connections, component values, and operating conditions. Start by identifying nodes and current paths.

Voltage or potential difference V Electrical energy difference per unit charge between two points. Unit: volt (V). Saying a node is at 5 V means 5 V relative to a reference.

Current I Rate of charge flow through a path. Unit: ampere (A). One ampere means one coulomb of charge passes per second.

Conventional current The direction positive charge would flow. In a metal, electrons move opposite this direction. Circuit current arrows usually use conventional current.

Charge Q An electrical property of particles, measured in coulombs (C). A capacitor stores separated positive and negative charge on its two sides.

Resistance R Opposition to current in an ohmic element. Unit: ohm (Ω). At the same voltage, a larger resistance gives less current.

Power P Rate of energy transfer or conversion, measured in watts (W). A resistor converts electrical energy into heat; its dissipation is P = VI.

Energy The amount transferred or stored, measured in joules (J). Power describes how fast energy transfers; energy describes how much transfers.

Node A set of points connected by ideal wire with the same voltage. A wire can bend or cross the drawing without changing which node it belongs to.

Branch A path between nodes containing one or more components. Current can split into separate branches at a node.

Loop A closed path that returns to its starting node. Use voltage rises and drops around a loop when applying KVL.

Ground or reference The point chosen as 0 V for the analysis or measurement. Circuit ground does not automatically mean a physical earth connection.

Series connection An unbranched path through components, so they carry the same current. Series resistor voltage drops add.

Parallel connection Components connected to the same two nodes, so they have the same voltage. Total current is the sum of the branch currents.

Open circuit A broken or disconnected current path. Ideal current through the break is zero, but voltage can exist across it.

Short circuit A very low resistance connection between points. The voltage across an ideal short is zero; the surrounding circuit determines the current.

Ohms Law For an ohmic resistor, V = IR. Use the voltage across and current through that same resistor, with consistent units.

Kirchhoffs Current Law KCL Current entering a node equals current leaving it. This is the current balance used to write node equations.

Kirchhoffs Voltage Law KVL The signed voltage changes around a closed loop sum to zero in the circuit models used here. Voltage rises balance voltage drops.

Definition of Terms Continued

Resistors capacitors and timing

Ideal component A perfect, simplified version used in a calculation. An ideal wire has 0 Ω resistance and no voltage drop; an ideal voltage source holds its stated voltage.

Voltage drop The reduction in voltage when moving across a component in a specified direction. For a resistor along current flow, drop = IR. An ideal wire drops 0 V.

Equivalent value The value of one component that can replace a group for the calculation at its two outside connections. Two 2 kΩ resistors in parallel are equivalent to one 1 kΩ resistor.

Resistor A component designed to provide resistance. It is used to limit current, divide voltage, or set operating conditions.

Nominal value and tolerance The nominal value is the stated value; tolerance is the allowed variation. A 1 kΩ resistor with ±5% tolerance can be 950 to 1050 Ω.

Power rating and derating The rating is allowable dissipation under specified conditions. Derating reduces that allowance, often at higher temperature. Compare calculated power with the datasheet limit.

Capacitor A component with two conductors separated by an insulating material. It stores energy in an electric field and carries current when its voltage changes.

Capacitance C Charge stored per volt for a linear capacitor: C = Q/V. Unit: farad (F). More capacitance means more charge at the same voltage. A 10 µF capacitor at 1 V has 10 µC of charge magnitude.

Farad F The unit of capacitance. One farad equals one coulomb per volt. Common smaller units are microfarads (µF), nanofarads (nF), and picofarads (pF).

Electric field and dielectric The electric field is associated with force on charge and stores capacitor energy. The dielectric is the insulating material between the capacitor’s conductors.

Electrolytic capacitor A capacitor type commonly used for larger capacitance. Most electrolytic capacitors are polarized, so their positive and negative terminals must be connected correctly.

Equivalent series resistance ESR A resistance used to model losses inside a real capacitor. High ESR can increase heating and reduce filtering performance.

Leakage current Small unwanted current through a component that ideally blocks it. Capacitor leakage means a real capacitor is not a perfect open circuit in steady DC.

Time constant τ The characteristic response time of a first order circuit. For a simple RC circuit, $\tau = RC$. Charging from zero reaches about 63.2% of final voltage after one τ.

Initial condition The circuit state before a change, such as the capacitor voltage just before a switch closes. It determines the starting point of the response.

Transient response The changing behavior after switching or another disturbance, while voltages and currents move toward their eventual behavior.

Steady state Behavior after the initial transient has settled. With constant DC, values become constant; AC steady state may continue changing in a repeating pattern.

Charging and discharging Charging builds stored charge and energy; discharging releases stored energy through a path. In a simple RC circuit, capacitor voltage changes exponentially.

exp and exponential change exp(x) means e raised to x, where e is about 2.718. $\exp(-t/\tau)$ describes a decaying quantity; at $t = \tau$ it equals about 0.368.

Definition of Terms Continued

Meters and changing signals

Digital multimeter DMM An instrument that can measure several quantities, commonly voltage, current, and resistance. The selected mode and lead jacks determine how it works.

Voltmeter Measures voltage between two points. Connect it in parallel. High input resistance reduces the current it draws from the circuit.

Ammeter Measures current through a path. Insert it in series. Its resistance and burden voltage can change the current being measured.

Ohmmeter Measures resistance using its own test stimulus. Connect across an unpowered component; discharge capacitors and account for alternate circuit paths.

Input resistance The resistance a connected instrument presents at its input in a stated mode. Finite voltmeter input resistance can change the voltage it measures.

Internal resistance Resistance inside a source or instrument, or its equivalent model. Source resistance causes voltage drop under load; ammeter resistance adds drop in series.

Burden voltage and shunt Burden voltage is the drop introduced by current measurement. A shunt is a known resistor that converts current to voltage for sensing: $V_{shunt} = I \times R_{shunt}$.

Load and loading effect A load draws current from a source or circuit node. Loading is the change caused by connecting that load, including a measuring instrument.

Continuity and OL Continuity indicates a conducting path below the meter's threshold. OL means overload or above range; in resistance mode it often indicates an open path.

Four wire or Kelvin measurement Uses two leads to force current and two to sense voltage across a resistor, reducing errors from lead and contact resistance.

Frequency f and period T Frequency is cycles per second, measured in hertz (Hz). Period is seconds per cycle. They are related by $T = 1/f$.

Peak and peak to peak Peak is the largest magnitude from zero. Peak to peak is maximum minus minimum; a sine wave from −5 V to +5 V is 10 V peak to peak.

Root mean square RMS An effective signal value. For voltage across a resistor, RMS gives the DC equivalent for average heating. For a pure sine with no offset, $V_{rms} = V_{peak}/\sqrt{2}$.

Reactance X The effect of energy storage on AC opposition in capacitors or inductors. Unit: ohm (Ω). Capacitive reactance magnitude is $|X_c| = 1/(2\pi fC)$.

Impedance Z The total voltage-to-current relationship for sinusoidal AC, including resistance, reactance, and phase. Unit: ohm (Ω). It is not generally just a real resistance.

Low pass filter and cutoff frequency A low pass filter reduces higher frequency signals. For a simple RC filter measured across C, cutoff $f_c = 1/(2\pi RC)$, where output amplitude is about 70.7% of input.

Definition of Terms Continued

Diodes models and practical limits

Diode A component with a nonlinear voltage-current relationship that conducts much more readily in its forward direction than in reverse below breakdown.

Anode A and cathode K The diode terminals. Forward conventional current flows from A to K. The line in the schematic symbol identifies K; many axial parts have a cathode band.

Forward bias The anode is at a higher voltage than the cathode, encouraging forward conduction. Forward current rises with applied voltage according to the diode's actual characteristic.

Reverse bias The anode is at a lower voltage than the cathode. A real diode usually carries small leakage current until sufficient reverse voltage produces breakdown.

Forward voltage drop VF The voltage across a diode while it conducts forward. It depends on current, temperature, and type. A 0.7 V silicon drop is an approximation.

Breakdown voltage The reverse voltage region where diode current rises sharply. Current must be limited. Zener diodes are designed for specified breakdown operation.

Rectification Converting an alternating input into a unidirectional output. A diode rectifier produces pulsating DC; a capacitor can reduce ripple.

Ripple A changing component superimposed on a DC level. For example, a rectifier and filter may leave a small periodic variation in the output voltage.

Light emitting diode LED A diode that emits light under forward current. It needs current limiting and an appropriate operating current; its forward voltage depends on the device.

Schottky diode A diode based on a metal-semiconductor junction. It often has lower forward drop than a silicon PN diode, with different leakage and voltage limits.

Zener diode A diode intended to operate at a specified reverse breakdown voltage with controlled current, often used for voltage references or limiting.

Ideal and constant drop diode models Simplifications for calculation. An ideal ON diode drops 0 V; a constant drop ON diode uses a stated VF. An OFF diode has zero current when breakdown is excluded.

Nonlinear component A component whose current does not scale directly with voltage over the full operating range. A diode cannot be replaced by one fixed resistance for all conditions.

Current limiting Controlling current to stay within an intended operating value. For a simple resistor and LED path, use R = (Vs − VF)/I and check component limits.

Voltage rating and datasheet A voltage rating states an allowed limit under specified conditions. The datasheet gives the part's ratings, characteristics, test conditions, and operating guidance.

Symbols that can be confusing

C can mean capacitance when it is a variable, or coulomb when it is a unit of charge. **V** can mean voltage or its unit, volt. **μ** means one millionth; **m** means one thousandth; **k** means one thousand; and **M** means one million. Capitalization matters: mΩ and MΩ are very different.

Σ means "add all the signed terms." **τ** is tau, the time constant. **dv/dt** means how quickly voltage changes with time. **||** means connected in parallel. Read these as words first, then use them in a formula.